INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

Advance Natural Language Processing

Project Team: LexFlow

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Team

Name	RollNumber
Shiva Shankar Gande	2023202005
Akshay Kohad	2023202007

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Introduction:

We started using Mbart 50 as our baseline model and baseline Dataset as Hinglish-TOP-Dataset and Dakshina Dataset but due to unavailability of enough compute resources we shifted and explored some smaller yet efficient models like T5 small, Transformer and GRU with Attention Mechanism and used some efficient datasets from hugging face to get decent results.

1.Model : t5_small

Dataset: findnitai/english-to-hinglish

It is an English to Hinglish Dataset aggregated from publicly available data sources.

Sources:

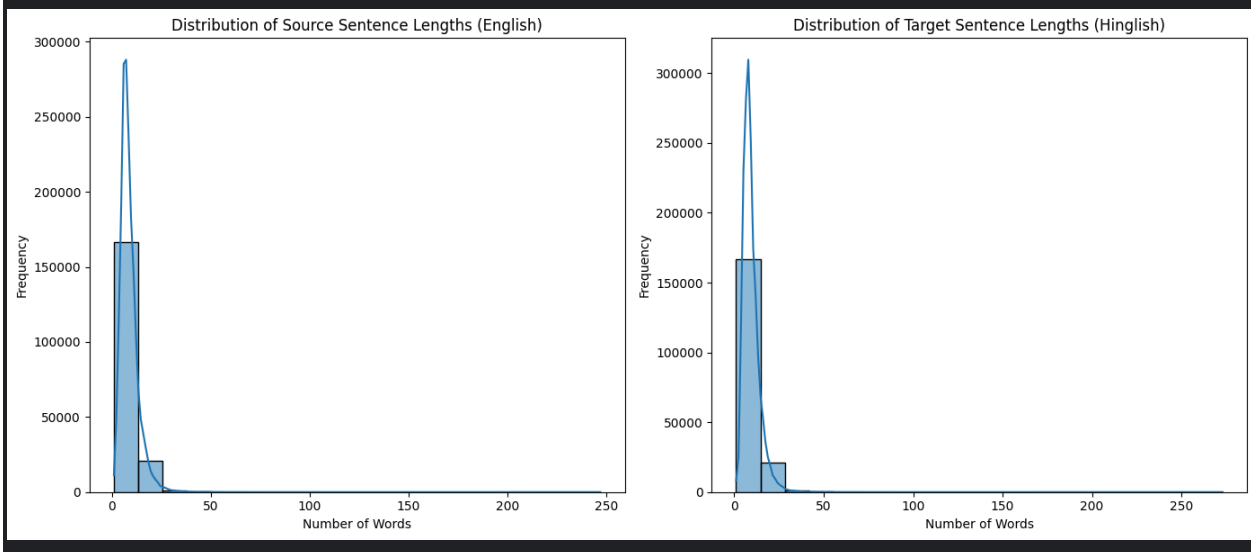
1. Hinglish TOP Dataset
2. CMU English Dog
3. HinGE
4. PHINC

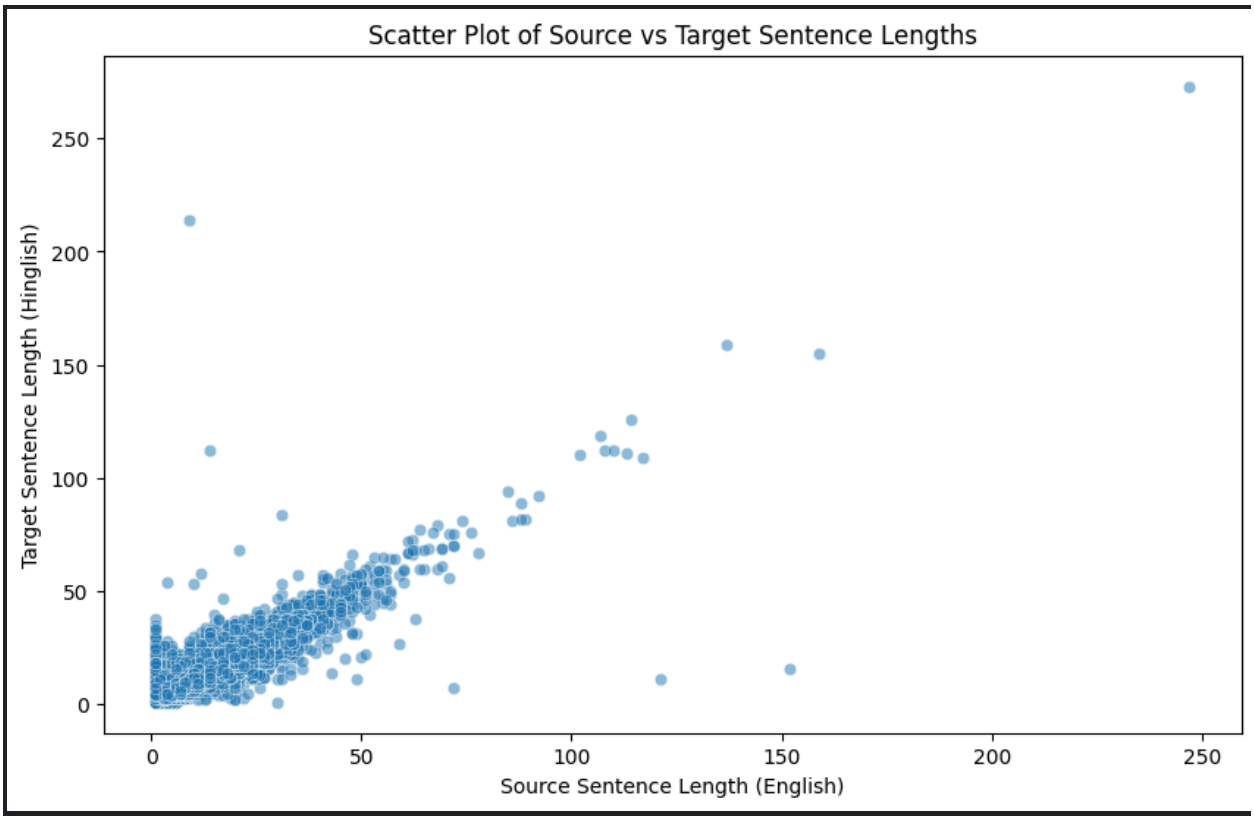
Exploratory analysis

source : 1 - Human Annotated , source : 0 - Synthetically Generated

```
translation
0
{'en': 'What's the name of the movie', 'hi_ng': 'film ka kya naam hai',
'source': 1}
1
{'en': 'Hi, the rotten tomatoes score is great but the meta critic score seems
a little low a movie of this quality. ', 'hi_ng': 'namaste, sada hua tomatoes
score mahaan hai, lekin meta critic score is gunavatta kee philm se thoda kam
lagata hai.', 'source': 1}
2
{'en': 'Do you think you will like the movie', 'hi_ng': 'kya aapako lagata hai
ki aapako film pasand aaegee', 'source': 1}
3
{'en': 'What kind of movie is it', 'hi_ng': 'yah kis tarah kee philm hai',
'source': 1}
4
{'en': 'when was the movie made?', 'hi_ng': 'film  kab banee thee?', 'source':
1}

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 189102 entries, 0 to 189101
Data columns (total 1 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   translation  189102 non-null  object
dtypes: object(1)
memory usage: 1.4+ MB

Number of Null values in each column:
translation    0
dtype: int64

Basic Statistics of Source Sentence Length:
count    189102.000000
mean          8.732271
std           4.744171
min           1.000000
25%           6.000000
50%           8.000000
75%          11.000000
max         247.000000
Name: source_length, dtype: float64
```

```
Basic Statistics of Target Sentence Length:
count    189102.000000
mean          9.398425
std           4.907075
min           1.000000
25%           6.000000
50%           8.000000
75%          11.000000
max         273.000000
Name: target_length, dtype: float64
```




Scatter Plot of Source vs Target Sentence Lengths
Target Sentence Length (Hinglish)
Source Sentence Length (English)
0
50
100
150
200
250

Word Cloud of Target Sentences (Hinglish)
kare
ko
ke
liye
kya
hai
alarm
mere
mujhe
set
tak
kitna
kab
wala
karo
karen
please
aaj
subah
aap
ki
baje
koi
ek
aur
message
mausam
yaad
timer
mera
mei
karne
kal
reminder
traffic
raat
Friday
Mujhe
aj
hu
band
nahi
kuch
hoon
har
Saturday
bajhe
se
time
weekend
subha
meri
mai
hoga
baarish
drive
work
event
add
aa
minute
weather
song
par
ka
kitni
music
sabhi
dilaye
paa
hone
cancel
mein
ghar
raha
sakte
kaisa
forecast
kar
Monday
New
samay
chahiye
play
agle
agar
toh
bajao
ho
karo
pm
kitne
temperature
dikhao
lagega
delete
kardo
pehle
shuru
ye
abhi

1.Training 30 percent of the dataset with 3 epochs

1.Parameters used

```
# ======================================
# Configuration Dictionary
# ======================================

config = {
    # ======================================
    # 1. Model and Tokenizer Settings
    # ======================================
    "model_name": "t5-small",  # Pre-trained model name
    "tokenizer_name": "t5-small",  # Tokenizer name (usually same as model)
    "source_lang": "en",  # Source language code
    "target_lang": "hi_ng",  # Target language code (Hinglish)
    "source_prefix": "translate English to Hinglish: ",  # Prefix for
source sentences

    # ======================================
    # 2. Tokenizer Training Parameters
    # ======================================
    "tokenizer_vocab_size": 32128,  # Vocabulary size for tokenizer
    "tokenizer_training_chunk_size": 500,  # Number of samples per chunk
during tokenizer training
    "tokenizer_file": "tokenizer_{0}.json",  # File pattern for saving
tokenizers

    # ======================================
    # 3. Data Loading and Sampling
    # ======================================
    "dataset_name": "findnitai/english-to-hinglish",  # Huggingface dataset
identifier
    "sample_fraction": 0.3,  # Fraction of the dataset to use (30%)
    "test_size": 0.1,  # Proportion of the dataset to include in the test
split (10%)
    "random_seed": 42,  # Seed for reproducibility

    # ======================================
    # 4. Preprocessing Parameters
    # ======================================
    "max_source_length": 128,  # Maximum token length for source sentences
    "max_target_length": 128,  # Maximum token length for target sentences
```

```
    "padding": "max_length",  # Padding strategy ("max_length", "longest",
or "do_not_pad")

    # ======================================
    # 5. Training Parameters
    # ======================================
    "num_epochs": 3,  # Number of training epochs
    "per_device_train_batch_size": 32,  # Training batch size per GPU/CPU
    "per_device_eval_batch_size": 32,  # Evaluation batch size per GPU/CPU
    "learning_rate": 5e-5,  # Learning rate for the optimizer
    "weight_decay": 0.0,  # Weight decay for the optimizer
    "num_warmup_steps": 0,  # Number of warmup steps for learning rate
scheduler
    "lr_scheduler_name": "linear",  # Learning rate scheduler type

    # ======================================
    # 6. Device and GPU Settings
    # ======================================
    "device": torch.device("cuda" if torch.cuda.is_available() else "cpu"),
# Computation device
    "num_gpus": torch.cuda.device_count(),  # Number of available GPUs

    # ======================================
    # 7. Data Collator and Dataloader Settings
    # ======================================
    "data_collator": "DataCollatorForSeq2Seq",  # Type of data collator
    "dataloader_num_workers": 4,  # Number of subprocesses for data loading

    # ======================================
    # 8. Optimization and Scheduler Settings
    # ======================================
    "optimizer_name": "AdamW",  # Optimizer type
    "optimizer_params": {
        "lr": 5e-5,
        "eps": 1e-8
    },
    "scheduler_params": {
        "name": "linear",
        "num_warmup_steps": 0
    },

    # ======================================
    # 9. Evaluation Metrics
```

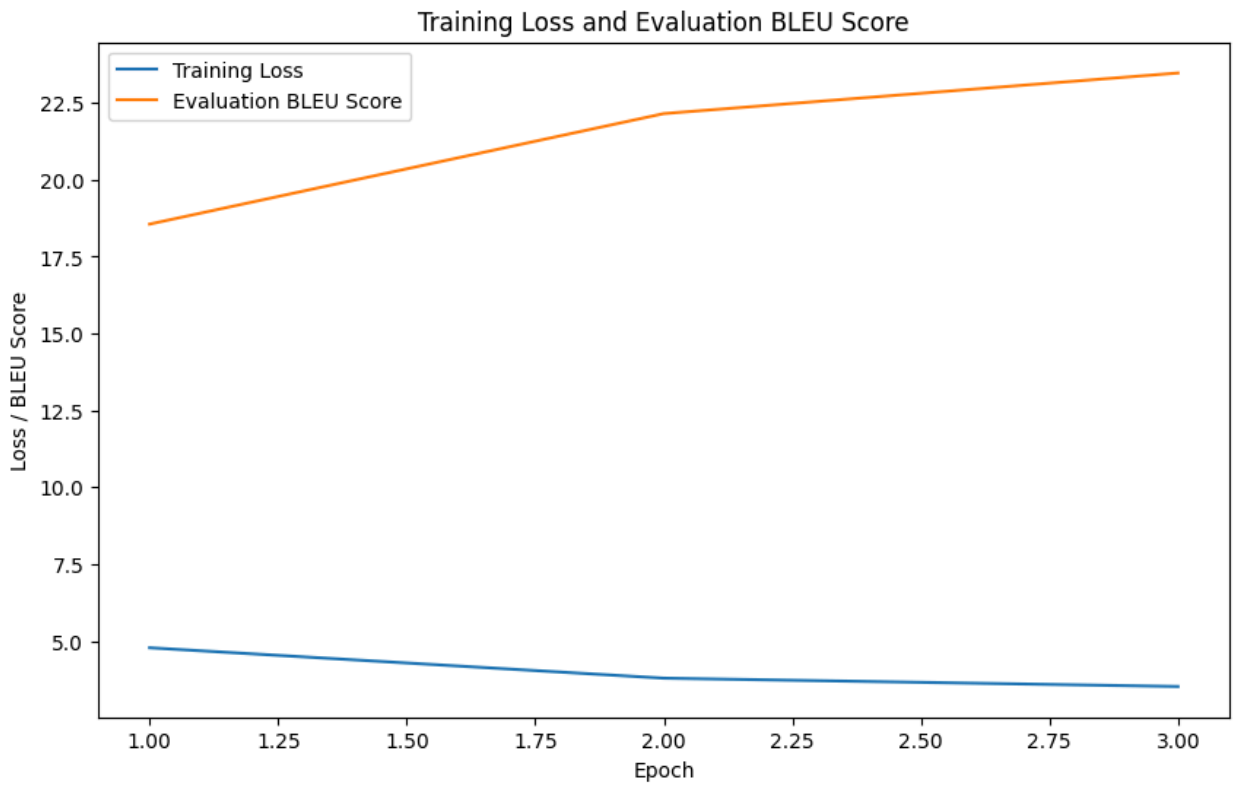
```
    # ======================================
    "metric_name": "sacrebleu",  # Evaluation metric
    "evaluation_steps": 1,  # Frequency of evaluation (every epoch)

    # ======================================
    # 10. Checkpointing and Saving
    # ======================================
    "model_folder": "weights",  # Directory to save model checkpoints
    "model_basename": "tmodel_",  # Basename for saved models
    "preload": None,  # Option to preload a model ('latest' or specific
epoch)

    # ======================================
    # 11. Miscellaneous
    # ======================================
    "experiment_name": "runs/tmodel",  # Experiment name for logging
    "logging_steps": 100,  # Frequency of logging training metrics
}
```

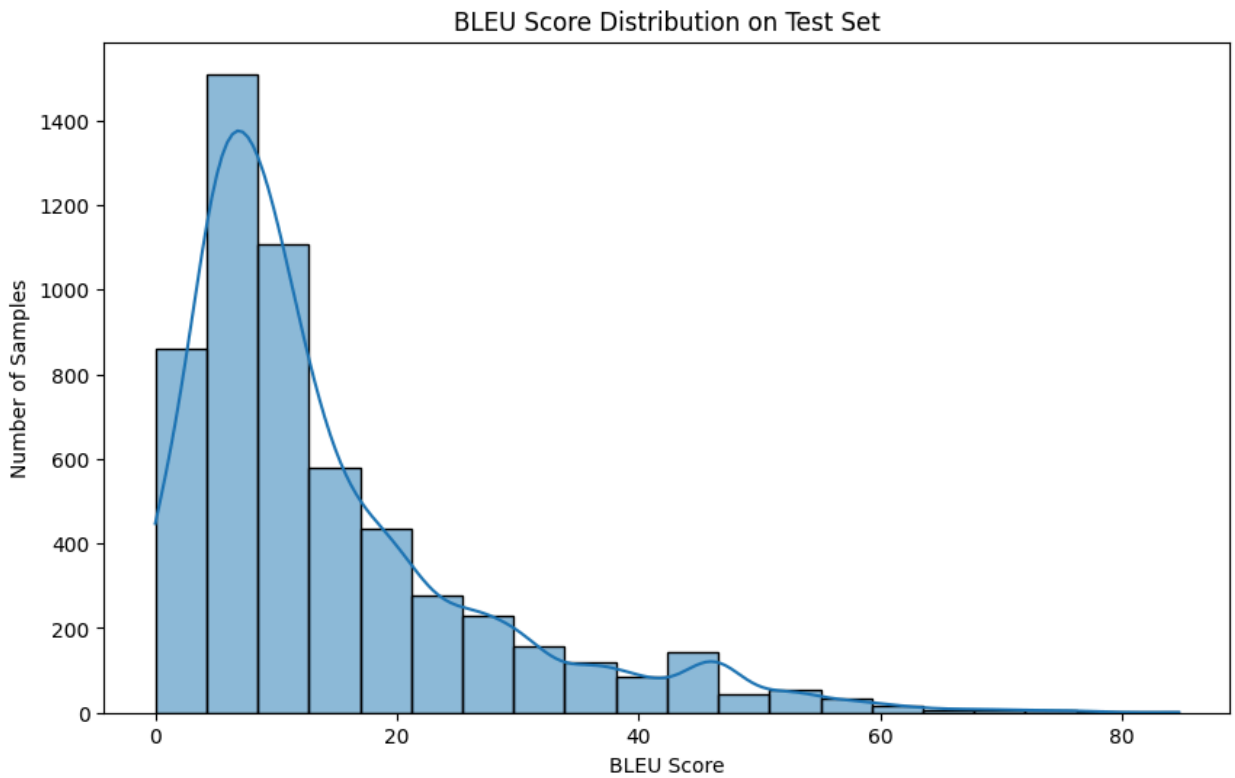
2.Training:

```
Using device: cuda
Number of GPUs available: 2
Using DataParallel for multi-GPU training.


Number of training samples: 51057
Number of evaluation samples: 5673
Total number of sentences in training data: 102114
Total number of sentences in evaluation data: 11346

Map: 100%
 51057/51057 [00:22<00:00, 2279.81 examples/s]
Map: 100%
 5673/5673 [00:02<00:00, 2051.04 examples/s]
Starting training...
Epoch [1/3], Step [0/1596], Loss: 6.8455
Epoch [1/3], Step [100/1596], Loss: 6.3996
Epoch [1/3], Step [200/1596], Loss: 6.0618
Epoch [1/3], Step [300/1596], Loss: 5.8546
```

```
Epoch [1/3], Step [400/1596], Loss: 5.6938
Epoch [1/3], Step [500/1596], Loss: 5.5690
Epoch [1/3], Step [600/1596], Loss: 5.4623
Epoch [1/3], Step [700/1596], Loss: 5.3562
Epoch [1/3], Step [800/1596], Loss: 5.2662
Epoch [1/3], Step [900/1596], Loss: 5.1840
Epoch [1/3], Step [1000/1596], Loss: 5.1183
Epoch [1/3], Step [1100/1596], Loss: 5.0458
Epoch [1/3], Step [1200/1596], Loss: 4.9832
Epoch [1/3], Step [1300/1596], Loss: 4.9231
Epoch [1/3], Step [1400/1596], Loss: 4.8733
Epoch [1/3], Step [1500/1596], Loss: 4.8266
Epoch [1/3], Evaluation BLEU Score: 18.56
Epoch [2/3], Step [0/1596], Loss: 4.7363
Epoch [2/3], Step [100/1596], Loss: 4.0437
Epoch [2/3], Step [200/1596], Loss: 4.0391
Epoch [2/3], Step [300/1596], Loss: 3.9982
Epoch [2/3], Step [400/1596], Loss: 3.9806
Epoch [2/3], Step [500/1596], Loss: 3.9668
Epoch [2/3], Step [600/1596], Loss: 3.9467
Epoch [2/3], Step [700/1596], Loss: 3.9276
Epoch [2/3], Step [800/1596], Loss: 3.9132
Epoch [2/3], Step [900/1596], Loss: 3.8990
Epoch [2/3], Step [1000/1596], Loss: 3.8862
Epoch [2/3], Step [1100/1596], Loss: 3.8675
Epoch [2/3], Step [1200/1596], Loss: 3.8498
Epoch [2/3], Step [1300/1596], Loss: 3.8313
Epoch [2/3], Step [1400/1596], Loss: 3.8155
Epoch [2/3], Step [1500/1596], Loss: 3.8073
Epoch [2/3], Evaluation BLEU Score: 22.15
Epoch [3/3], Step [0/1596], Loss: 4.1915
Epoch [3/3], Step [100/1596], Loss: 3.6109
Epoch [3/3], Step [200/1596], Loss: 3.5843
Epoch [3/3], Step [300/1596], Loss: 3.5598
Epoch [3/3], Step [400/1596], Loss: 3.5539
Epoch [3/3], Step [500/1596], Loss: 3.5563
Epoch [3/3], Step [600/1596], Loss: 3.5595
Epoch [3/3], Step [700/1596], Loss: 3.5478
Epoch [3/3], Step [800/1596], Loss: 3.5392
Epoch [3/3], Step [900/1596], Loss: 3.5321
Epoch [3/3], Step [1000/1596], Loss: 3.5297
Epoch [3/3], Step [1100/1596], Loss: 3.5324
Epoch [3/3], Step [1200/1596], Loss: 3.5356
Epoch [3/3], Step [1300/1596], Loss: 3.5279
```

```
Epoch [3/3], Step [1400/1596], Loss: 3.5271
Epoch [3/3], Step [1500/1596], Loss: 3.5266
Epoch [3/3], Evaluation BLEU Score: 23.47
Training completed.
```





3.Testing:

```
Average BLEU score on test set: 14.16

Sample translations from the test set:

Source: Can I go out with just a jacket today?
Reference: Kya aaj mujhe sirf jacket pehen kar bahar nikalna chahiye?
Prediction: kya kya just ko jacket ke liye I go out kya hai?

Source: play piano music
Reference: piano music play karo
Prediction: piano music play kare

Source: Turn my alarms off now
Reference: Mere alarms ko aab band kare
Prediction: mere alarms ko band karo

Source: Loki's a likable villain I'd say.
Reference: Main bolunga k Loki ek likable villain hai
Prediction: Loki's a likable villain I'd say.
```

```
Source: Tell me the traffic at 5 pm
Reference: Mujhe sham 5 bajhe traffic ke baare me batao
Prediction: mujhe 5 pm ko traffic kaisa hai
```

```
Custom Translation:
Input: I was waiting for my bag
Translated Output: mere bag ke liye was waiting kya hai
```

2.Training the whole dataset with 5 epochs

Trained on a dataset of 189,102 sentences, the model was split into training, development, and test sets. The training process ran for 5 hours using two Tesla T4 GPUs.

1.Parameters used

```
# ======================================
# Configuration Dictionary
# ======================================

config = {
    # ======================================
    # 1. Model and Tokenizer Settings
    # ======================================
    "model_name": "t5-small",  # Pre-trained model name
    "tokenizer_name": "t5-small",  # Tokenizer name (usually same as model)
    "source_lang": "en",  # Source language code
    "target_lang": "hi_ng",  # Target language code (Hinglish)
    "source_prefix": "translate English to Hinglish: ",  # Prefix for
source sentences

    # ======================================
    # 2. Data Loading and Sampling
    # ======================================
    "dataset_name": "findnitai/english-to-hinglish",  # Huggingface dataset
identifier
    "sample_fraction": 1.0,  # Fraction of the dataset to use (100%)
    "test_size": 0.1,  # Proportion of the dataset to include in the test
split (10%)
    "random_seed": 42,  # Seed for reproducibility
```

```
    # ========================================
    # 3. Preprocessing Parameters
    # ========================================
    "max_source_length": 128,  # Maximum token length for source sentences
    "max_target_length": 128,  # Maximum token length for target sentences
    "padding": "max_length",  # Padding strategy ("max_length", "longest",
or "do_not_pad")

    # ========================================
    # 4. Training Parameters
    # ========================================
    "num_epochs": 5,  # Number of training epochs
    "per_device_train_batch_size": 16,  # Training batch size per GPU/CPU
    "per_device_eval_batch_size": 32,  # Evaluation batch size per GPU/CPU
    "learning_rate": 3e-5,  # Learning rate for the optimizer
    "weight_decay": 0.0,  # Weight decay for the optimizer
    "num_warmup_steps": 0,  # Number of warmup steps for learning rate
scheduler
    "lr_scheduler_name": "linear",  # Learning rate scheduler type
    "gradient_accumulation_steps": 2,  # Number of steps to accumulate
gradients

    # ========================================
    # 5. Device and GPU Settings
    # ========================================
    "device": torch.device("cuda" if torch.cuda.is_available() else "cpu"),
# Computation device
    "num_gpus": torch.cuda.device_count(),  # Number of available GPUs

    # ========================================
    # 6. Data Collator and Dataloader Settings
    # ========================================
    "data_collator": "DataCollatorForSeq2Seq",  # Type of data collator
    "dataloader_num_workers": 4,  # Number of subprocesses for data loading

    # ========================================
    # 7. Optimization and Scheduler Settings
    # ========================================
    "optimizer_name": "AdamW",  # Optimizer type
    "optimizer_params": {
        "lr": 3e-5,
        "eps": 1e-8
```

```
    },
    "scheduler_params": {
        "name": "linear",
        "num_warmup_steps": 0
    },

    # ======================================
    # 8. Evaluation Metrics
    # ======================================
    "metric_name": "sacrebleu",  # Evaluation metric
    "evaluation_steps": 1,  # Frequency of evaluation (every epoch)

    # ======================================
    # 9. Checkpointing and Saving
    # ======================================
    "model_folder": "weights",  # Directory to save model checkpoints
    "model_basename": "tmodel_",  # Basename for saved models
    "preload": None,  # Option to preload a model ('latest' or specific
epoch)

    # ======================================
    # 10. Miscellaneous
    # ======================================
    "experiment_name": "runs/tmodel",  # Experiment name for logging
    "logging_steps": 100,  # Frequency of logging training metrics
}
```

2.Training

```
Using device: cuda
Number of GPUs available: 2
Using DataParallel for multi-GPU training.
Number of training samples: 170191
Number of evaluation samples: 18911
Total number of sentences in training data: 340382
Total number of sentences in evaluation data: 37822

Map: 100%
 170191/170191 [01:15<00:00, 2293.32 examples/s]
Map: 100%
 18911/18911 [00:08<00:00, 2340.50 examples/s]
```

```
Starting training...

Epoch [1/5], Step [0/10637], Loss: 4.9462
Epoch [1/5], Step [100/10637], Loss: 4.7178
Epoch [1/5], Step [200/10637], Loss: 4.7198
Epoch [1/5], Step [300/10637], Loss: 4.7011
Epoch [1/5], Step [400/10637], Loss: 4.6676
Epoch [1/5], Step [500/10637], Loss: 4.6191
Epoch [1/5], Step [600/10637], Loss: 4.5696
Epoch [1/5], Step [700/10637], Loss: 4.5184
Epoch [1/5], Step [800/10637], Loss: 4.4672
Epoch [1/5], Step [900/10637], Loss: 4.4117
Epoch [1/5], Step [1000/10637], Loss: 4.3528
Epoch [1/5], Step [1100/10637], Loss: 4.2935
Epoch [1/5], Step [1200/10637], Loss: 4.2312
Epoch [1/5], Step [1300/10637], Loss: 4.1704
Epoch [1/5], Step [1400/10637], Loss: 4.1120
Epoch [1/5], Step [1500/10637], Loss: 4.0497
Epoch [1/5], Step [1600/10637], Loss: 3.9924
Epoch [1/5], Step [1700/10637], Loss: 3.9357
Epoch [1/5], Step [1800/10637], Loss: 3.8802
Epoch [1/5], Step [1900/10637], Loss: 3.8270
Epoch [1/5], Step [2000/10637], Loss: 3.7743
Epoch [1/5], Step [2100/10637], Loss: 3.7250
Epoch [1/5], Step [2200/10637], Loss: 3.6769
Epoch [1/5], Step [2300/10637], Loss: 3.6312
Epoch [1/5], Step [2400/10637], Loss: 3.5886
Epoch [1/5], Step [2500/10637], Loss: 3.5470
Epoch [1/5], Step [2600/10637], Loss: 3.5070
Epoch [1/5], Step [2700/10637], Loss: 3.4719
Epoch [1/5], Step [2800/10637], Loss: 3.4353
Epoch [1/5], Step [2900/10637], Loss: 3.3999
Epoch [1/5], Step [3000/10637], Loss: 3.3648
Epoch [1/5], Step [3100/10637], Loss: 3.3313
Epoch [1/5], Step [3200/10637], Loss: 3.2993
Epoch [1/5], Step [3300/10637], Loss: 3.2675
Epoch [1/5], Step [3400/10637], Loss: 3.2376
Epoch [1/5], Step [3500/10637], Loss: 3.2075
Epoch [1/5], Step [3600/10637], Loss: 3.1795
Epoch [1/5], Step [3700/10637], Loss: 3.1503
Epoch [1/5], Step [3800/10637], Loss: 3.1227
Epoch [1/5], Step [3900/10637], Loss: 3.0957
Epoch [1/5], Step [4000/10637], Loss: 3.0677
Epoch [1/5], Step [4100/10637], Loss: 3.0412
```

```
Epoch [1/5], Step [4200/10637], Loss: 3.0160
Epoch [1/5], Step [4300/10637], Loss: 2.9917
Epoch [1/5], Step [4400/10637], Loss: 2.9672
Epoch [1/5], Step [4500/10637], Loss: 2.9433
Epoch [1/5], Step [4600/10637], Loss: 2.9213
Epoch [1/5], Step [4700/10637], Loss: 2.8980
Epoch [1/5], Step [4800/10637], Loss: 2.8754
Epoch [1/5], Step [4900/10637], Loss: 2.8535
Epoch [1/5], Step [5000/10637], Loss: 2.8328
Epoch [1/5], Step [5100/10637], Loss: 2.8125
Epoch [1/5], Step [5200/10637], Loss: 2.7916
Epoch [1/5], Step [5300/10637], Loss: 2.7707
Epoch [1/5], Step [5400/10637], Loss: 2.7518
Epoch [1/5], Step [5500/10637], Loss: 2.7319
Epoch [1/5], Step [5600/10637], Loss: 2.7128
Epoch [1/5], Step [5700/10637], Loss: 2.6945
Epoch [1/5], Step [5800/10637], Loss: 2.6762
Epoch [1/5], Step [5900/10637], Loss: 2.6584
Epoch [1/5], Step [6000/10637], Loss: 2.6408
Epoch [1/5], Step [6100/10637], Loss: 2.6233
Epoch [1/5], Step [6200/10637], Loss: 2.6066
Epoch [1/5], Step [6300/10637], Loss: 2.5901
Epoch [1/5], Step [6400/10637], Loss: 2.5737
Epoch [1/5], Step [6500/10637], Loss: 2.5573
Epoch [1/5], Step [6600/10637], Loss: 2.5410
Epoch [1/5], Step [6700/10637], Loss: 2.5257
Epoch [1/5], Step [6800/10637], Loss: 2.5103
Epoch [1/5], Step [6900/10637], Loss: 2.4952
Epoch [1/5], Step [7000/10637], Loss: 2.4803
Epoch [1/5], Step [7100/10637], Loss: 2.4658
Epoch [1/5], Step [7200/10637], Loss: 2.4517
Epoch [1/5], Step [7300/10637], Loss: 2.4378
Epoch [1/5], Step [7400/10637], Loss: 2.4242
Epoch [1/5], Step [7500/10637], Loss: 2.4111
Epoch [1/5], Step [7600/10637], Loss: 2.3974
Epoch [1/5], Step [7700/10637], Loss: 2.3842
Epoch [1/5], Step [7800/10637], Loss: 2.3717
Epoch [1/5], Step [7900/10637], Loss: 2.3587
Epoch [1/5], Step [8000/10637], Loss: 2.3465
Epoch [1/5], Step [8100/10637], Loss: 2.3341
Epoch [1/5], Step [8200/10637], Loss: 2.3225
Epoch [1/5], Step [8300/10637], Loss: 2.3112
Epoch [1/5], Step [8400/10637], Loss: 2.2995
Epoch [1/5], Step [8500/10637], Loss: 2.2871
Epoch [1/5], Step [8600/10637], Loss: 2.2751
```

```
Epoch [1/5], Step [8700/10637], Loss: 2.2643
Epoch [1/5], Step [8800/10637], Loss: 2.2530
Epoch [1/5], Step [8900/10637], Loss: 2.2423
Epoch [1/5], Step [9000/10637], Loss: 2.2318
Epoch [1/5], Step [9100/10637], Loss: 2.2210
Epoch [1/5], Step [9200/10637], Loss: 2.2103
Epoch [1/5], Step [9300/10637], Loss: 2.1997
Epoch [1/5], Step [9400/10637], Loss: 2.1896
Epoch [1/5], Step [9500/10637], Loss: 2.1795
Epoch [1/5], Step [9600/10637], Loss: 2.1692
Epoch [1/5], Step [9700/10637], Loss: 2.1601
Epoch [1/5], Step [9800/10637], Loss: 2.1504
Epoch [1/5], Step [9900/10637], Loss: 2.1414
Epoch [1/5], Step [10000/10637], Loss: 2.1321
Epoch [1/5], Step [10100/10637], Loss: 2.1229
Epoch [1/5], Step [10200/10637], Loss: 2.1135
Epoch [1/5], Step [10300/10637], Loss: 2.1044
Epoch [1/5], Step [10400/10637], Loss: 2.0960
Epoch [1/5], Step [10500/10637], Loss: 2.0875
Epoch [1/5], Step [10600/10637], Loss: 2.0790
Epoch [1/5], Evaluation BLEU Score: 36.91
Epoch [2/5], Step [0/10637], Loss: 1.2476
Epoch [2/5], Step [100/10637], Loss: 1.2060
Epoch [2/5], Step [200/10637], Loss: 1.1979
Epoch [2/5], Step [300/10637], Loss: 1.1930
Epoch [2/5], Step [400/10637], Loss: 1.1922
Epoch [2/5], Step [500/10637], Loss: 1.1871
Epoch [2/5], Step [600/10637], Loss: 1.1818
Epoch [2/5], Step [700/10637], Loss: 1.1801
Epoch [2/5], Step [800/10637], Loss: 1.1822
Epoch [2/5], Step [900/10637], Loss: 1.1819
Epoch [2/5], Step [1000/10637], Loss: 1.1804
Epoch [2/5], Step [1100/10637], Loss: 1.1787
Epoch [2/5], Step [1200/10637], Loss: 1.1734
Epoch [2/5], Step [1300/10637], Loss: 1.1706
Epoch [2/5], Step [1400/10637], Loss: 1.1686
Epoch [2/5], Step [1500/10637], Loss: 1.1628
Epoch [2/5], Step [1600/10637], Loss: 1.1608
Epoch [2/5], Step [1700/10637], Loss: 1.1577
Epoch [2/5], Step [1800/10637], Loss: 1.1565
Epoch [2/5], Step [1900/10637], Loss: 1.1525
Epoch [2/5], Step [2000/10637], Loss: 1.1503
Epoch [2/5], Step [2100/10637], Loss: 1.1501
Epoch [2/5], Step [2200/10637], Loss: 1.1463
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Epoch [2/5], Step [2300/10637], Loss: 1.1439
Epoch [2/5], Step [2400/10637], Loss: 1.1422
Epoch [2/5], Step [2500/10637], Loss: 1.1408
Epoch [2/5], Step [2600/10637], Loss: 1.1393
Epoch [2/5], Step [2700/10637], Loss: 1.1382
Epoch [2/5], Step [2800/10637], Loss: 1.1376
Epoch [2/5], Step [2900/10637], Loss: 1.1357
Epoch [2/5], Step [3000/10637], Loss: 1.1324
Epoch [2/5], Step [3100/10637], Loss: 1.1307
Epoch [2/5], Step [3200/10637], Loss: 1.1300
Epoch [2/5], Step [3300/10637], Loss: 1.1286
Epoch [2/5], Step [3400/10637], Loss: 1.1261
Epoch [2/5], Step [3500/10637], Loss: 1.1240
Epoch [2/5], Step [3600/10637], Loss: 1.1216
Epoch [2/5], Step [3700/10637], Loss: 1.1191
Epoch [2/5], Step [3800/10637], Loss: 1.1171
Epoch [2/5], Step [3900/10637], Loss: 1.1151
Epoch [2/5], Step [4000/10637], Loss: 1.1138
Epoch [2/5], Step [4100/10637], Loss: 1.1112
Epoch [2/5], Step [4200/10637], Loss: 1.1103
Epoch [2/5], Step [4300/10637], Loss: 1.1097
Epoch [2/5], Step [4400/10637], Loss: 1.1081
Epoch [2/5], Step [4500/10637], Loss: 1.1064
Epoch [2/5], Step [4600/10637], Loss: 1.1050
Epoch [2/5], Step [4700/10637], Loss: 1.1032
Epoch [2/5], Step [4800/10637], Loss: 1.1020
Epoch [2/5], Step [4900/10637], Loss: 1.1006
Epoch [2/5], Step [5000/10637], Loss: 1.0996
Epoch [2/5], Step [5100/10637], Loss: 1.0971
Epoch [2/5], Step [5200/10637], Loss: 1.0960
Epoch [2/5], Step [5300/10637], Loss: 1.0947
Epoch [2/5], Step [5400/10637], Loss: 1.0936
Epoch [2/5], Step [5500/10637], Loss: 1.0934
Epoch [2/5], Step [5600/10637], Loss: 1.0922
Epoch [2/5], Step [5700/10637], Loss: 1.0906
Epoch [2/5], Step [5800/10637], Loss: 1.0896
Epoch [2/5], Step [5900/10637], Loss: 1.0877
Epoch [2/5], Step [6000/10637], Loss: 1.0861
Epoch [2/5], Step [6100/10637], Loss: 1.0849
Epoch [2/5], Step [6200/10637], Loss: 1.0841
Epoch [2/5], Step [6300/10637], Loss: 1.0832
Epoch [2/5], Step [6400/10637], Loss: 1.0816
Epoch [2/5], Step [6500/10637], Loss: 1.0799
Epoch [2/5], Step [6600/10637], Loss: 1.0790
Epoch [2/5], Step [6700/10637], Loss: 1.0773
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Epoch [2/5], Step [6800/10637], Loss: 1.0765
Epoch [2/5], Step [6900/10637], Loss: 1.0754
Epoch [2/5], Step [7000/10637], Loss: 1.0746
Epoch [2/5], Step [7100/10637], Loss: 1.0728
Epoch [2/5], Step [7200/10637], Loss: 1.0722
Epoch [2/5], Step [7300/10637], Loss: 1.0711
Epoch [2/5], Step [7400/10637], Loss: 1.0700
Epoch [2/5], Step [7500/10637], Loss: 1.0691
Epoch [2/5], Step [7600/10637], Loss: 1.0684
Epoch [2/5], Step [7700/10637], Loss: 1.0669
Epoch [2/5], Step [7800/10637], Loss: 1.0654
Epoch [2/5], Step [7900/10637], Loss: 1.0643
Epoch [2/5], Step [8000/10637], Loss: 1.0637
Epoch [2/5], Step [8100/10637], Loss: 1.0624
Epoch [2/5], Step [8200/10637], Loss: 1.0613
Epoch [2/5], Step [8300/10637], Loss: 1.0600
Epoch [2/5], Step [8400/10637], Loss: 1.0587
Epoch [2/5], Step [8500/10637], Loss: 1.0581
Epoch [2/5], Step [8600/10637], Loss: 1.0567
Epoch [2/5], Step [8700/10637], Loss: 1.0556
Epoch [2/5], Step [8800/10637], Loss: 1.0547
Epoch [2/5], Step [8900/10637], Loss: 1.0537
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Epoch [2/5], Step [9100/10637], Loss: 1.0513
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Epoch [2/5], Step [9400/10637], Loss: 1.0474
Epoch [2/5], Step [9500/10637], Loss: 1.0459
Epoch [2/5], Step [9600/10637], Loss: 1.0452
Epoch [2/5], Step [9700/10637], Loss: 1.0436
Epoch [2/5], Step [9800/10637], Loss: 1.0430
Epoch [2/5], Step [9900/10637], Loss: 1.0420
Epoch [2/5], Step [10000/10637], Loss: 1.0410
Epoch [2/5], Step [10100/10637], Loss: 1.0397
Epoch [2/5], Step [10200/10637], Loss: 1.0381
Epoch [2/5], Step [10300/10637], Loss: 1.0370
Epoch [2/5], Step [10400/10637], Loss: 1.0361
Epoch [2/5], Step [10500/10637], Loss: 1.0356
Epoch [2/5], Step [10600/10637], Loss: 1.0344
Epoch [2/5], Evaluation BLEU Score: 44.20
Epoch [3/5], Step [0/10637], Loss: 0.7940
Epoch [3/5], Step [100/10637], Loss: 0.9388
Epoch [3/5], Step [200/10637], Loss: 0.9321
Epoch [3/5], Step [300/10637], Loss: 0.9335
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Epoch [3/5], Step [400/10637], Loss: 0.9309
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Epoch [3/5], Step [600/10637], Loss: 0.9253
Epoch [3/5], Step [700/10637], Loss: 0.9262
Epoch [3/5], Step [800/10637], Loss: 0.9187
Epoch [3/5], Step [900/10637], Loss: 0.9138
Epoch [3/5], Step [1000/10637], Loss: 0.9151
Epoch [3/5], Step [1100/10637], Loss: 0.9162
Epoch [3/5], Step [1200/10637], Loss: 0.9154
Epoch [3/5], Step [1300/10637], Loss: 0.9169
Epoch [3/5], Step [1400/10637], Loss: 0.9165
Epoch [3/5], Step [1500/10637], Loss: 0.9155
Epoch [3/5], Step [1600/10637], Loss: 0.9145
Epoch [3/5], Step [1700/10637], Loss: 0.9129
Epoch [3/5], Step [1800/10637], Loss: 0.9108
Epoch [3/5], Step [1900/10637], Loss: 0.9091
Epoch [3/5], Step [2000/10637], Loss: 0.9103
Epoch [3/5], Step [2100/10637], Loss: 0.9095
Epoch [3/5], Step [2200/10637], Loss: 0.9090
Epoch [3/5], Step [2300/10637], Loss: 0.9066
Epoch [3/5], Step [2400/10637], Loss: 0.9080
Epoch [3/5], Step [2500/10637], Loss: 0.9072
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Epoch [3/5], Step [2800/10637], Loss: 0.9030
Epoch [3/5], Step [2900/10637], Loss: 0.9021
Epoch [3/5], Step [3000/10637], Loss: 0.9023
Epoch [3/5], Step [3100/10637], Loss: 0.9012
Epoch [3/5], Step [3200/10637], Loss: 0.9016
Epoch [3/5], Step [3300/10637], Loss: 0.9025
Epoch [3/5], Step [3400/10637], Loss: 0.9025
Epoch [3/5], Step [3500/10637], Loss: 0.9021
Epoch [3/5], Step [3600/10637], Loss: 0.9029
Epoch [3/5], Step [3700/10637], Loss: 0.9026
Epoch [3/5], Step [3800/10637], Loss: 0.9015
Epoch [3/5], Step [3900/10637], Loss: 0.9010
Epoch [3/5], Step [4000/10637], Loss: 0.9005
Epoch [3/5], Step [4100/10637], Loss: 0.9002
Epoch [3/5], Step [4200/10637], Loss: 0.9005
Epoch [3/5], Step [4300/10637], Loss: 0.8994
Epoch [3/5], Step [4400/10637], Loss: 0.8992
Epoch [3/5], Step [4500/10637], Loss: 0.8979
Epoch [3/5], Step [4600/10637], Loss: 0.8981
Epoch [3/5], Step [4700/10637], Loss: 0.8978
Epoch [3/5], Step [4800/10637], Loss: 0.8976
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Epoch [3/5], Step [4900/10637], Loss: 0.8965
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Epoch [3/5], Step [5100/10637], Loss: 0.8950
Epoch [3/5], Step [5200/10637], Loss: 0.8957
Epoch [3/5], Step [5300/10637], Loss: 0.8955
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Epoch [3/5], Step [5700/10637], Loss: 0.8939
Epoch [3/5], Step [5800/10637], Loss: 0.8930
Epoch [3/5], Step [5900/10637], Loss: 0.8925
Epoch [3/5], Step [6000/10637], Loss: 0.8921
Epoch [3/5], Step [6100/10637], Loss: 0.8915
Epoch [3/5], Step [6200/10637], Loss: 0.8909
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Epoch [3/5], Step [6600/10637], Loss: 0.8890
Epoch [3/5], Step [6700/10637], Loss: 0.8892
Epoch [3/5], Step [6800/10637], Loss: 0.8886
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Epoch [3/5], Step [7000/10637], Loss: 0.8877
Epoch [3/5], Step [7100/10637], Loss: 0.8872
Epoch [3/5], Step [7200/10637], Loss: 0.8858
Epoch [3/5], Step [7300/10637], Loss: 0.8853
Epoch [3/5], Step [7400/10637], Loss: 0.8853
Epoch [3/5], Step [7500/10637], Loss: 0.8847
Epoch [3/5], Step [7600/10637], Loss: 0.8844
Epoch [3/5], Step [7700/10637], Loss: 0.8842
Epoch [3/5], Step [7800/10637], Loss: 0.8837
Epoch [3/5], Step [7900/10637], Loss: 0.8833
Epoch [3/5], Step [8000/10637], Loss: 0.8833
Epoch [3/5], Step [8100/10637], Loss: 0.8831
Epoch [3/5], Step [8200/10637], Loss: 0.8828
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Epoch [3/5], Step [8400/10637], Loss: 0.8827
Epoch [3/5], Step [8500/10637], Loss: 0.8822
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Epoch [3/5], Step [8800/10637], Loss: 0.8816
Epoch [3/5], Step [8900/10637], Loss: 0.8817
Epoch [3/5], Step [9000/10637], Loss: 0.8812
Epoch [3/5], Step [9100/10637], Loss: 0.8809
Epoch [3/5], Step [9200/10637], Loss: 0.8807
Epoch [3/5], Step [9300/10637], Loss: 0.8802
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Epoch [3/5], Step [9400/10637], Loss: 0.8796
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Epoch [3/5], Step [9800/10637], Loss: 0.8777
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Epoch [3/5], Step [10300/10637], Loss: 0.8759
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Epoch [3/5], Step [10500/10637], Loss: 0.8755
Epoch [3/5], Step [10600/10637], Loss: 0.8752
Epoch [3/5], Evaluation BLEU Score: 46.48
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Epoch [4/5], Step [200/10637], Loss: 0.8389
Epoch [4/5], Step [300/10637], Loss: 0.8368
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Epoch [4/5], Step [1400/10637], Loss: 0.8271
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Epoch [4/5], Step [2700/10637], Loss: 0.8271
Epoch [4/5], Step [2800/10637], Loss: 0.8277
Epoch [4/5], Step [2900/10637], Loss: 0.8266
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Epoch [4/5], Step [3000/10637], Loss: 0.8256
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Epoch [4/5], Step [3200/10637], Loss: 0.8249
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Epoch [4/5], Step [4000/10637], Loss: 0.8212
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Epoch [4/5], Step [4300/10637], Loss: 0.8205
Epoch [4/5], Step [4400/10637], Loss: 0.8212
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Epoch [4/5], Step [4800/10637], Loss: 0.8202
Epoch [4/5], Step [4900/10637], Loss: 0.8201
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Epoch [4/5], Step [7000/10637], Loss: 0.8156
Epoch [4/5], Step [7100/10637], Loss: 0.8154
Epoch [4/5], Step [7200/10637], Loss: 0.8149
Epoch [4/5], Step [7300/10637], Loss: 0.8147
Epoch [4/5], Step [7400/10637], Loss: 0.8144
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Epoch [4/5], Step [7500/10637], Loss: 0.8145
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Epoch [4/5], Step [7800/10637], Loss: 0.8151
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Epoch [4/5], Step [10500/10637], Loss: 0.8109
Epoch [4/5], Step [10600/10637], Loss: 0.8109
Epoch [4/5], Evaluation BLEU Score: 47.75
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Epoch [5/5], Step [800/10637], Loss: 0.8028
Epoch [5/5], Step [900/10637], Loss: 0.7999
Epoch [5/5], Step [1000/10637], Loss: 0.7956
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Epoch [5/5], Step [1400/10637], Loss: 0.7967
Epoch [5/5], Step [1500/10637], Loss: 0.7959
Epoch [5/5], Step [1600/10637], Loss: 0.7963
Epoch [5/5], Step [1700/10637], Loss: 0.7958
Epoch [5/5], Step [1800/10637], Loss: 0.7960
Epoch [5/5], Step [1900/10637], Loss: 0.7971
Epoch [5/5], Step [2000/10637], Loss: 0.7951
Epoch [5/5], Step [2100/10637], Loss: 0.7946
Epoch [5/5], Step [2200/10637], Loss: 0.7956
Epoch [5/5], Step [2300/10637], Loss: 0.7948
Epoch [5/5], Step [2400/10637], Loss: 0.7955
Epoch [5/5], Step [2500/10637], Loss: 0.7962
Epoch [5/5], Step [2600/10637], Loss: 0.7956
Epoch [5/5], Step [2700/10637], Loss: 0.7961
Epoch [5/5], Step [2800/10637], Loss: 0.7954
Epoch [5/5], Step [2900/10637], Loss: 0.7954
Epoch [5/5], Step [3000/10637], Loss: 0.7954
Epoch [5/5], Step [3100/10637], Loss: 0.7962
Epoch [5/5], Step [3200/10637], Loss: 0.7960
Epoch [5/5], Step [3300/10637], Loss: 0.7954
Epoch [5/5], Step [3400/10637], Loss: 0.7954
Epoch [5/5], Step [3500/10637], Loss: 0.7946
Epoch [5/5], Step [3600/10637], Loss: 0.7933
Epoch [5/5], Step [3700/10637], Loss: 0.7933
Epoch [5/5], Step [3800/10637], Loss: 0.7916
Epoch [5/5], Step [3900/10637], Loss: 0.7917
Epoch [5/5], Step [4000/10637], Loss: 0.7911
Epoch [5/5], Step [4100/10637], Loss: 0.7910
Epoch [5/5], Step [4200/10637], Loss: 0.7912
Epoch [5/5], Step [4300/10637], Loss: 0.7907
Epoch [5/5], Step [4400/10637], Loss: 0.7900
Epoch [5/5], Step [4500/10637], Loss: 0.7881
Epoch [5/5], Step [4600/10637], Loss: 0.7887
Epoch [5/5], Step [4700/10637], Loss: 0.7884
Epoch [5/5], Step [4800/10637], Loss: 0.7876
Epoch [5/5], Step [4900/10637], Loss: 0.7882
Epoch [5/5], Step [5000/10637], Loss: 0.7888
Epoch [5/5], Step [5100/10637], Loss: 0.7892
Epoch [5/5], Step [5200/10637], Loss: 0.7889
Epoch [5/5], Step [5300/10637], Loss: 0.7895
Epoch [5/5], Step [5400/10637], Loss: 0.7893
Epoch [5/5], Step [5500/10637], Loss: 0.7891
```
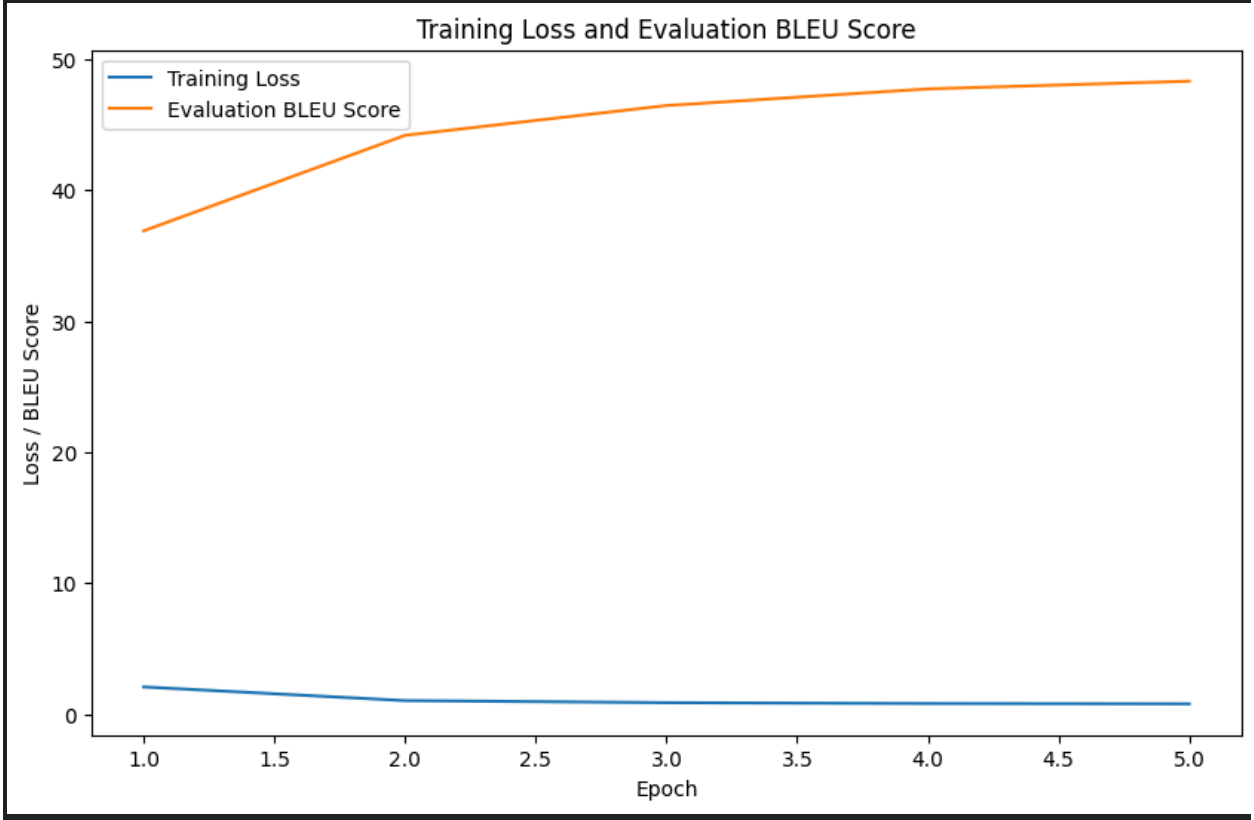

```
Epoch [5/5], Step [5600/10637], Loss: 0.7893
Epoch [5/5], Step [5700/10637], Loss: 0.7890
Epoch [5/5], Step [5800/10637], Loss: 0.7887
Epoch [5/5], Step [5900/10637], Loss: 0.7889
Epoch [5/5], Step [6000/10637], Loss: 0.7889
Epoch [5/5], Step [6100/10637], Loss: 0.7890
Epoch [5/5], Step [6200/10637], Loss: 0.7892
Epoch [5/5], Step [6300/10637], Loss: 0.7885
Epoch [5/5], Step [6400/10637], Loss: 0.7880
Epoch [5/5], Step [6500/10637], Loss: 0.7872
Epoch [5/5], Step [6600/10637], Loss: 0.7871
Epoch [5/5], Step [6700/10637], Loss: 0.7876
Epoch [5/5], Step [6800/10637], Loss: 0.7873
Epoch [5/5], Step [6900/10637], Loss: 0.7873
Epoch [5/5], Step [7000/10637], Loss: 0.7866
Epoch [5/5], Step [7100/10637], Loss: 0.7864
Epoch [5/5], Step [7200/10637], Loss: 0.7863
Epoch [5/5], Step [7300/10637], Loss: 0.7863
Epoch [5/5], Step [7400/10637], Loss: 0.7866
Epoch [5/5], Step [7500/10637], Loss: 0.7862
Epoch [5/5], Step [7600/10637], Loss: 0.7859
Epoch [5/5], Step [7700/10637], Loss: 0.7862
Epoch [5/5], Step [7800/10637], Loss: 0.7865
Epoch [5/5], Step [7900/10637], Loss: 0.7861
Epoch [5/5], Step [8000/10637], Loss: 0.7858
Epoch [5/5], Step [8100/10637], Loss: 0.7861
Epoch [5/5], Step [8200/10637], Loss: 0.7854
Epoch [5/5], Step [8300/10637], Loss: 0.7851
Epoch [5/5], Step [8400/10637], Loss: 0.7849
Epoch [5/5], Step [8500/10637], Loss: 0.7848
Epoch [5/5], Step [8600/10637], Loss: 0.7852
Epoch [5/5], Step [8700/10637], Loss: 0.7850
Epoch [5/5], Step [8800/10637], Loss: 0.7852
Epoch [5/5], Step [8900/10637], Loss: 0.7846
Epoch [5/5], Step [9000/10637], Loss: 0.7846
Epoch [5/5], Step [9100/10637], Loss: 0.7846
Epoch [5/5], Step [9200/10637], Loss: 0.7842
Epoch [5/5], Step [9300/10637], Loss: 0.7838
Epoch [5/5], Step [9400/10637], Loss: 0.7835
Epoch [5/5], Step [9500/10637], Loss: 0.7831
Epoch [5/5], Step [9600/10637], Loss: 0.7831
Epoch [5/5], Step [9700/10637], Loss: 0.7826
Epoch [5/5], Step [9800/10637], Loss: 0.7831
Epoch [5/5], Step [9900/10637], Loss: 0.7830
Epoch [5/5], Step [10000/10637], Loss: 0.7831
```

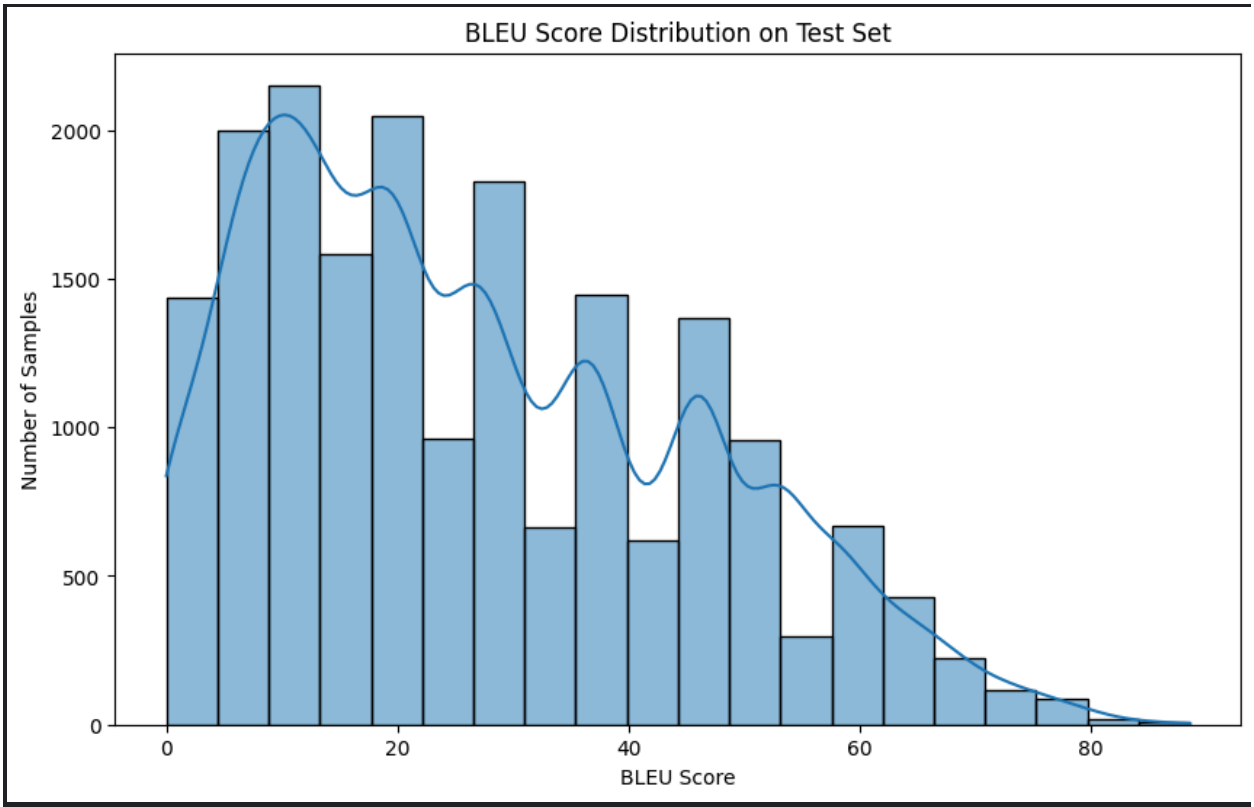
```
Epoch [5/5], Step [10100/10637], Loss: 0.7825
Epoch [5/5], Step [10200/10637], Loss: 0.7826
Epoch [5/5], Step [10300/10637], Loss: 0.7828
Epoch [5/5], Step [10400/10637], Loss: 0.7829
Epoch [5/5], Step [10500/10637], Loss: 0.7829
Epoch [5/5], Step [10600/10637], Loss: 0.7833
Epoch [5/5], Evaluation BLEU Score: 48.34
Training completed.
```




```
Generating predictions on the test set...
```

```
Computing BLEU scores for each test sample...
```

3.Testing




```
Average BLEU score on test set: 26.98
```

```
Sample translations from the test set:

Source: delete all recurring alarms
Reference: sarey recurring alarms delete karo
Prediction: sabhi recurring alarms delete kare

Source: what is the best route to the racetrack from my sister's house, if i
left around 2 pm?
Reference: agar mai 2 pm ke aas paas nikal jaoon to meri sister ke ghar se
racetrack tak pahunchne ke liye sabse best route konsa hai?
Prediction: agar mai 2 pm ko nikalta hoon toh mere sister ke ghar se racetrack
tak sabse acha route kya hai?

Source: play the top 40 songs right now
Reference: top 40 songs ko abhi play kare
Prediction: abhi top 40 songs bajao

Source: i want to cancel that alarm
```

```
Reference: mai wo alarm cancel karna chahta hoon
Prediction: mai ye alarm cancel karna chahta hoon

Source: alert me at 6 tomorrow instead of 7 am.
Reference: mujhe kal 7 am ke bajaye 6 ko alert kare
Prediction: mujhe kal subah 6 baje subah 7 baje ke liye alert kare
```

```
custom_sentence = "I was waiting for my bag"
translated_sentence = translate_sentence(custom_sentence)
print("\nCustom Translation:")
print(f"Input: {custom_sentence}")
print(f"Translated Output: {translated_sentence}")
```

```

Custom Translation:
Input: I was waiting for my bag
Translated Output: mujhe mere bag ke liye waiting karne ke liye
```

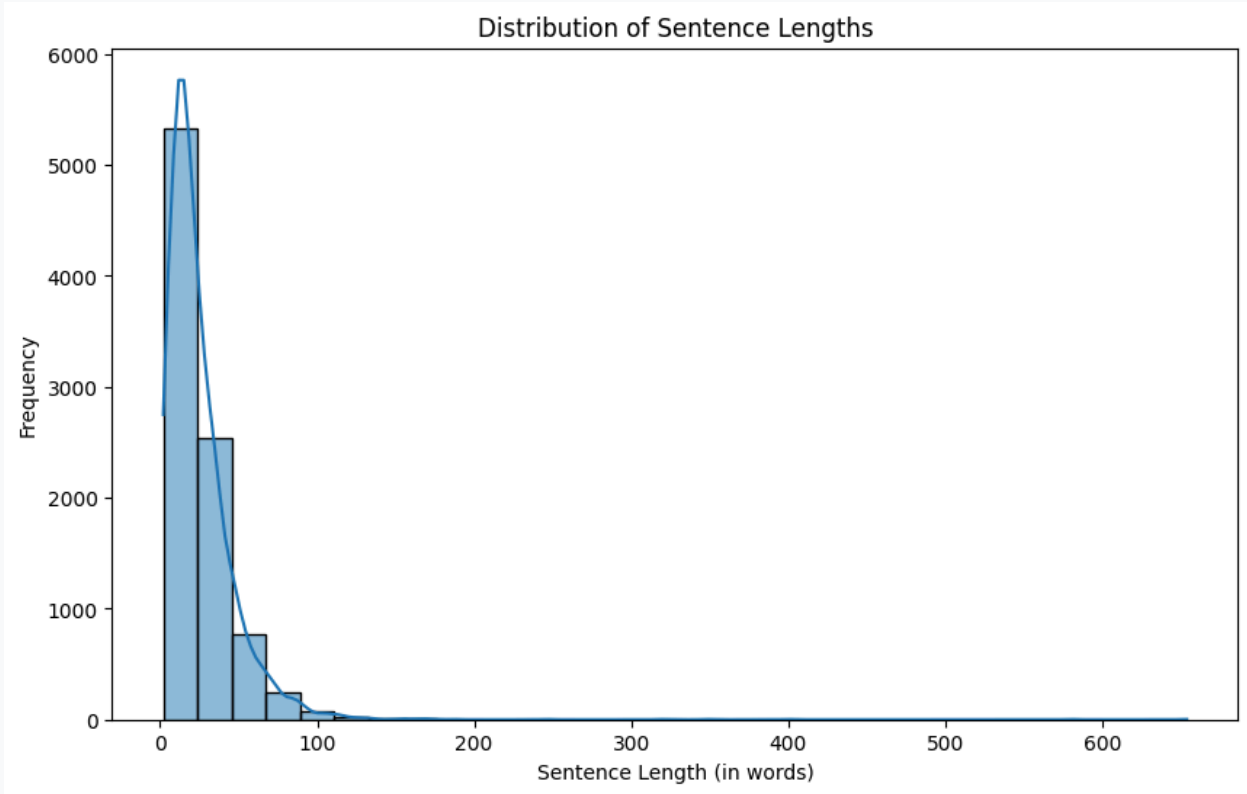
```
custom_sentence = "i go to school"
translated_sentence = translate_sentence(custom_sentence)
print("\nCustom Translation:")
print(f"Input: {custom_sentence}")
print(f"Translated Output: {translated_sentence}")
```

```
Custom Translation:
Input: i go to school
Translated Output: mujhe school par jaana chahiye
```

2.Model : Transformer

Dataset:[English-Hinglish Dataset]

Exploratory analysis



Number of unique words in the dataset: 12667

It is an English to Hinglish Dataset in devnagari script.
Dataset is divided into:
a) Train.txt -> Training Data consists of 7k Lines of English Sentences along with the target sentence in Hinglish separated by '\t' .
b) Dev.txt -> -> Validation Data consists of 1k Lines of English Sentences along with the target sentence in Hinglish separated by '\t' .
c) Test.txt -> -> Testing Data consists of 900 Lines of English Sentences along with the target sentence in Hinglish separated by '\t' .

1)Parameters Used

```
import torch
import torch.nn as nn
```

```python
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
from torch.nn.utils.rnn import pad_sequence
import math
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path

# Initialize empty lists for train, test, and dev data
train_data = []
test_data = []
dev_data = []

# Function to read data from a file and split into lists
def read_data(filename, data_list):
    with open(filename, 'r') as file:
        for line in file:
            sentence1, sentence2 = line.strip().split('\t')
            data_list.append((sentence1.lower(), sentence2.lower()))

# File paths for your data files
train_file = '/kaggle/input/codemix-dataset/Dataset 1 (English -
Hinglish)/train.txt'
test_file = '/kaggle/input/codemix-dataset/Dataset 1 (English -
Hinglish)/test.txt'
dev_file = '/kaggle/input/codemix-dataset/Dataset 1 (English -
Hinglish)/dev.txt'

# Read data into the lists
read_data(train_file, train_data)
read_data(test_file, test_data)
read_data(dev_file, dev_data)

# Print the first few entries of each list
print("Train Data:")
print(train_data[:5])  # Print the first 5 entries as an example

print("Test Data:")
print(test_data[:5])

print("Dev Data:")
```

```
print(dev_data[:5])




train_set = []
test_set = []
dev_set = []
for i in range(len(train_data)):
    if len(train_data[i][0].split()) > 60 or len(train_data[i][1].split())
> 60:
        continue
    train_set.append({'en':train_data[i][0],'fr': train_data[i][1]})
for i in range(len(test_data)):
    if len(test_data[i][0].split()) > 60 or len(test_data[i][1].split()) >
60:
        continue
    test_set.append({'en':test_data[i][0],'fr': test_data[i][1]})
for i in range(len(dev_data)):
    if len(dev_data[i][0].split()) > 60 or len(dev_data[i][1].split()) >
60:
        continue
    dev_set.append({'en':dev_data[i][0],'fr': dev_data[i][1]})
```

```
# warnings.filterwarnings("ignore")
config = {
    "batch_size": 8,
    "num_epochs": 10,
    "lr": 10**-4,
    "seq_len": 128,
    "d_model": 512,
    "datasource": [train_set,test_set],
    "lang_src": "en",
    "lang_tgt": "fr",
    "model_folder": "weights",
    "model_basename": "tmodel_",
    "preload": None,
    "tokenizer_file": "tokenizer_{0}.json",
    "experiment_name": "runs/tmodel"
```

```
}
```

2) Training

We have Train the model for 10 epochs with following output and loss values

```
Using device: cuda
Max length of source sentence: 64
Max length of target sentence: 63
No model to preload, starting from scratch

Processing Epoch 00: 100%|██████████| 877/877 [00:30<00:00, 29.09it/s,
loss=6.190]

--------------------------------------------------------------------------------
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं नहीं हूँ , यह movie के लिए , मैं नहीं है
--------------------------------------------------------------------------------

Processing Epoch 01: 100%|██████████| 877/877 [00:29<00:00, 29.91it/s,
loss=5.326]

--------------------------------------------------------------------------------
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
```

```
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं नहीं देखा , मैं देखा हूँ , मैं के लिए यह एक है
--------------------------------------------------------------------------------
-

Processing Epoch 02: 100%|██████████| 877/877 [00:29<00:00, 29.79it/s,
loss=4.027]

--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं नहीं देखा हूँ , मैं इस movie को करने के लिए नहीं कर रहा हूँ
--------------------------------------------------------------------------------
-

Processing Epoch 03: 100%|██████████| 877/877 [00:29<00:00, 29.91it/s,
loss=3.306]

--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं इस movie को नहीं देखा हूँ , तो मैं हूँ
--------------------------------------------------------------------------------
-
Processing Epoch 04: 100%|██████████| 877/877 [00:29<00:00, 29.93it/s,
```

```
loss=2.673]
--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं तो मैं इस movie को देखा हूँ तो मैं हूँ
--------------------------------------------------------------------------------
-
Processing Epoch 05: 100%|██████████| 877/877 [00:29<00:00, 29.74it/s,
loss=2.204]
--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , मैं इस movie को नहीं देखा हूँ , तो मैं जरूर हूँ
--------------------------------------------------------------------------------
-
Processing Epoch 06: 100%|██████████| 877/877 [00:29<00:00, 29.85it/s,
loss=1.989]

--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
```

```
PREDICTED: hello , मैं there , मैं यह movie नहीं देखा हूँ , मैं शायद मैं शायद खुद को देखने
के लिए
--------------------------------------------------------------------------------
-
Processing Epoch 07: 100%|██████████| 877/877 [00:29<00:00, 29.65it/s,
loss=1.521]

--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello there , मैं यह movie नहीं देखा हूँ , so मैं शायद मैं शायद हूँ इस तरह से
देखने के लिए
--------------------------------------------------------------------------------
-
Processing Epoch 08: 100%|██████████| 877/877 [00:29<00:00, 29.82it/s,
loss=1.473]

--------------------------------------------------------------------------------
-
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello , there , मैं यह movie नहीं देखा हूँ तो मैं एक minute minute की कोशिश
करता हूँ ये मेरे लिए
--------------------------------------------------------------------------------
-
Processing Epoch 09: 100%|██████████| 877/877 [00:29<00:00, 29.82it/s,
loss=1.407]

--------------------------------------------------------------------------------
-
```

```
   SOURCE: hello
   TARGET: hello
PREDICTED: hello
--------------------------------------------------------------------------------
-
   SOURCE: hello there , i have not seen this movie so im going to take a
minute to look it over :)
   TARGET: हैलो यार , मैं इस movie को नहीं देख हों तो , तो मैं थोड़ी देर के लिए इसको देख
लूंगा
PREDICTED: hello there , मैं यह movie नहीं देखा हूँ तो मैं minute करने के लिए एक minute
बाद में :)
```

3) Testing

We then Translate the English Sentence into Hinglish Sentence for all Dataset i.e Train,Dev and Test and calculate the resulting bleu score.

Train Dataset

```
SOURCE: the deal where the winner gets the loser's robot seems a bit extreme
PREDICTED: वह deal था कि deal robot को थोड़ा extreme था थोड़ा extreme था थोड़ा
extreme था
SOURCE: lol . i see the movie was filmed totally in michigan . good for them .
tax credits .
PREDICTED: lol , मैं michigan पूरा movie पूरा देखा है , tax credits अच्छा है tax
credits
SOURCE: this is more of an action film with robots
PREDICTED: यह एक action film से action film के साथ ही है
SOURCE: rotten tomatoes gives it 59 percent . i guess that means they say it's
an ok kind of move . i remember the boxing robots though .
PREDICTED: rotten tomatoes 59 59 59 59 % . मुझे लगता है कि यह कहता है , मुझे लगता
है . में boxing robots को याद है .
```

```
Average BLEU Score: 20.71
```

Validation Dataset

```
SOURCE: ok you think we have enough then .
PREDICTED: ok तुम हम लगता है फिर फिर से हम फिर लगता है .  Using device: cuda
   SOURCE: ok .
PREDICTED: ठीक है  Using device: cuda
   SOURCE: ok bro
PREDICTED: ठीक ठीक है  Using device: cuda
   SOURCE: shall we continue ?
PREDICTED: क्या हम जारी रख सकते हैं ?
   SOURCE: do you like we can
PREDICTED: तुम्हें लगता हम कर सकते हैं
   SOURCE: yeah i have fifteen thanks .
PREDICTED: हाँ मेरे पास  धन्यवाद धन्यवाद
```

```
Average BLEU Score: 9.39
```

Test Dataset

```
SOURCE: sounds like a good movie . can we finish now ?
PREDICTED: sound अच्छा movie है . हम finish conversation finish करने चाहिए ?

SOURCE: yep . thanks for chatting
PREDICTED: हां . thanks आपसे chat करने के लिए thanks

SOURCE: thanks , i will watch it . sounds good
PREDICTED: धन्यवाद मैं watch करूँगा .

SOURCE: hi !
PREDICTED: है !

SOURCE: did you like the movie despicable me ?
PREDICTED: क्या तुम्हें movie पसंद despicable despicable me ?

SOURCE: i did not realize this movie came out in 2010 , that seemed so long ago
.
PREDICTED: मुझे नहीं पता movie नहीं ये 2010 2010 में सबसे 2010 की सबसे long है .

Average BLEU Score: 7.98
```

Comparative Analysis of Models and Dataset

Model Used	Epochs	Dataset Used	Bleu Score
T5_small	5	findnitai/english-to-hinglish(Full)	26.98
T5_small	3	findnitai/english-to-hinglish(30%)	14.16
Transformer	10	English-to-Hinglish	7.98